Slide 1

IntellMeet

AI-Powered Enterprise Meeting & Collaboration Platform

Production-Grade Full-Stack MERN Application with Real-Time Video, AI Meeting Intelligence & Team Collaboration

MERN | WebRTC | P2P | Socket.io | GPT-4 | Live Deployed

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Problem & Solution

Problem

- Expensive SFU
- high cost
- no AI

Solution

- MERN + P2P WebRTC UDP SRTP
- zero cost
- 90% saving
- GPT-4

Scalable • Cost-Efficient • AI-Powered

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System Architecture (6-Layer)

1. Presentation Layer
- User interface & web/mobile clients
- API endpoints & dashboard
- Handles user interactions and display

2. Application Layer
- Request routing & orchestration
- Session management & authentication
- Coordinates business workflows

3. Service Layer
- Core services & microservices
- Business rules engine
- Integration with external APIs

4. Business Logic Layer
- Data processing & validation
- Calculations, rules, and domain logic
- Enforces policies and business rules

5. Data Access Layer
- Database connectors & ORM
- Caching layer (Redis)
- Data repositories and queries

6. Infrastructure Layer
- Cloud infrastructure & hosting
- Monitoring, logging, and security
- Networking, storage, and backups

Scalable • Secure • Modular • Fault-Tolerant

Slide 4 |

Real-time & Deployment

WebSocket Lifecycle

- **Connect:** Establish persistent WebSocket connection
- **Subscribe:** Client subscribes to event channels/topics
- **Stream:** Real-time bidirectional data flow
- **Reconnect:** Exponential backoff on disconnect
- **Close:** Graceful termination & cleanup

Deployment

- **Containerized:** Docker container with healthchecks
- **Autoscale:** Horizontal scaling based on connection count
- **Logging:** Centralized logs & metrics via Prometheus
- **CI/CD:** Automated deploy on merge to main

• Latency target: < 50ms • Secure: WSS + token auth • Monitored: real-time dashboards

Database & Security

Data Encryption

- AES-256 encryption for data at rest & in transit
- End-to-end encrypted backups
- Key management via KMS/HSM integration

Access Control

- Role-based access control (RBAC) enforced
- Multi-factor authentication (MFA) required
- Audit logs for all privilege changes

Backup & Recovery

- Automated daily backups with 30-day retention
- Point-in-time recovery for critical data
- Geo-redundant storage across regions

Threat Monitoring

- Real-time anomaly detection & alerts
- Automated vulnerability scanning weekly
- SIEM integration for incident response

Live Demo Flow

01 **Initialize Environment**
Launch container • Verify dependencies • Check GPU/CPU resources

02 **Load Data & Model**
Ingest sample dataset • Load pre-trained model • Validate input schema

03 **Run Inference**
Execute prediction pipeline • Stream results in real-time • Monitor latency & throughput

04 **Visualize Results**
Render charts & metrics • Highlight anomalies • Update dashboard live

05 **Export Output**
Save to cloud storage • Generate report PDF • Notify stakeholders

Conclusion

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- Reinforces the key findings, confirming improved efficiency and reduced latency by 30%
- Validates the framework's scalability across distributed environments
- Demonstrates robust performance under high-load conditions with consistent reliability

Future Scope

- Expansion to real-time adaptive learning modules for dynamic optimization
- Integration with edge computing to support low-latency IoT deployments
- Exploration of cross-domain applications in healthcare and autonomous systems

Thank You